Project Report On

RETAIL SALES ANALYSIS & BUSINESS INTELLIGENCE DASHBOARD

Project Type	Data Analytics & Business Intelligence
Skills	Python, Pandas, SQL, Power BI, DAX
Data Source	Retail Sales Transaction Dataset (CSV)
Dashboard	Interactive Power BI Dashboard
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This report documents the complete analytics workflow from raw retail transaction data to SQL-based business analysis and an interactive Power BI dashboard.

1. Problem Statement

Retail businesses generate large volumes of transaction data containing information about customers, products, quantities, prices, and sales amounts. However, raw transaction data is difficult to interpret directly and does not provide management with an efficient way to monitor business performance.

This project addresses the need for a centralized analytical solution that can identify revenue performance, product-category contribution, customer purchasing patterns, sales trends, and average transaction value. The goal is to transform raw retail transaction data into meaningful business insights using Python, SQL, MySQL, and Power BI.

2. Objectives

- Clean and validate the raw retail transaction dataset.
- Perform Exploratory Data Analysis (EDA) to understand sales and customer patterns.
- Store structured data in a MySQL database.
- Use SQL queries to calculate important business metrics.
- Create reusable SQL views for analytical reporting.
- Connect MySQL with Power BI.
- Develop an interactive Business Intelligence dashboard.
- Analyze revenue, products, customers, quantity, and sales trends.
- Generate actionable business insights and recommendations.

3. Dataset

The project uses a retail transaction-level dataset named **retail_sales.csv**. The dataset contains transaction, customer, product, quantity, pricing, and sales-value information.

Column	Description
Transaction ID	Unique identifier for each transaction
Date	Date of purchase
Customer ID	Unique customer identifier
Gender	Customer gender
Age	Customer age
Product Category	Category of purchased product
Quantity	Number of units purchased
Price per Unit	Price of one unit
Total Amount	Total value of the transaction

4. Methodology

4.1 Data Collection

The retail transaction dataset was collected in CSV format and used as the initial source of analysis.

4.2 Data Loading

The dataset was loaded using Python and Pandas. Initial inspection included viewing records, checking data types, descriptive statistics, missing values, and duplicate records.

4.3 Data Cleaning

The data was checked for missing values, duplicate records, invalid values, and appropriate data types.

The cleaned dataset was prepared for structured database storage and analysis.

4.4 Exploratory Data Analysis

EDA was performed to understand revenue distribution, product-category performance, customer demographics, quantity sold, sales trends, and transaction values. Pandas, NumPy, and visualization tools were used where applicable.

4.5 MySQL Database

The prepared data was stored in a MySQL database named **retail_sales**, with the main transaction table named **sales**. MySQL provides structured storage and a reliable environment for SQL-based analysis.

4.6 SQL Business Analysis

SQL was used to calculate total revenue, total orders, total quantity, average order value, revenue by product category, revenue by gender, monthly revenue, and category-wise quantity sold.

4.7 SQL Views

Reusable analytical views can be created for summarized metrics such as category revenue, gender revenue, monthly revenue, and sales summaries. These views simplify repeated reporting logic.

4.8 Power BI Dashboard

Power BI was connected to the MySQL database to build an interactive dashboard. KPI cards, charts, slicers, and DAX measures were used to present business performance in a user-friendly format.

5. Insights

The final numerical findings should be taken directly from the completed dashboard and SQL results. The analysis is designed to identify the following business insights:

- The highest-revenue product categories and their contribution to overall sales.
- The categories with the highest sales volume based on quantity sold.
- Customer purchasing patterns across gender and age groups.
- Monthly periods with relatively higher or lower revenue.

- Average revenue generated per transaction.
- Differences between high-volume categories and high-value categories.

Note: Replace this section with the actual numerical findings from the completed Power BI dashboard before submitting the report.

6. Business Recommendations

- Prioritize high-performing product categories through appropriate inventory planning and promotionalactivity.
- Investigate low-performing categories to understand whether pricing, product assortment, or marketing requires improvement.
- Use customer demographic patterns to support targeted marketing campaigns.
- Use monthly sales trends to improve inventory and promotional planning.
- Consider bundles, cross-selling, and promotional offers to improve average order value.

7. Technology Stack

Technology	Purpose
Python	Data preparation and analysis
Pandas	Data manipulation and preprocessing
NumPy	Numerical analysis
Matplotlib	Exploratory visualization
Jupyter Notebook	EDA and analysis documentation
MySQL	Structured data storage
SQL	Business analysis and aggregations
MySQL Workbench	Database management and query execution
Power BI	Interactive dashboard and visualization
DAX	Dynamic measures and KPI calculations
Git/GitHub	Version control and portfolio management

8. Conclusion

The Retail Sales Analysis project demonstrates an end-to-end Data Analytics and Business Intelligence workflow, starting from raw transaction data and progressing through data cleaning, exploratory analysis, database storage, SQL analysis, and Power BI visualization.

Python was used for data preparation and exploratory analysis, MySQL and SQL were used for structured storage and business analysis, and Power BI was used to transform analytical results into an interactive dashboard. The solution provides a centralized view of sales performance, product categories, customer behavior, and sales trends, supporting more informed business decisions.